Power BI Engineering Playbook

END-TO-END ETL, STAR SCHEMA MODELING, & DAX ARCHITECTURE

1. Architectural Overview & Data Strategy

Unstructured Excel spreadsheets often combine wide-pivoted layouts, embedded sub-totals, average calculations, and multiple distinct metric groupings across horizontal columns. To build an enterprise-grade, scalable Power BI model from such data, you must transform wide "human-readable" layouts into a normalized **Star Schema** "machine-readable" structure.

TARGET STAR SCHEMA ARCHITECTURE

```
┌──────────────────────────────────────┐
│              DateTable               │
│  (Dimension - DAX Generated Table)   │
├──────────────────────────────────────┤
│  Date (PK)                           │
│  Year                                │
│  Month Number                        │
│  Month                               │
│  Year Month                          │
└──────────────────┬───────────────────┘
                   │
                 1 │ (One)
                   │
                   │
                 * │ (Many)
┌──────────────────┴───────────────────┐
│              FactSales               │
│         (Consolidated Fact)          │
├──────────────────────────────────────┤
│  Salesperson                         │
│  Month                               │
│  Sales                               │
│  Profit                              │
│  MergeKey (Surrogate Composite Key)  │
│  Month Number                        │
│  Date (FK)                           │
└──────────────────────────────────────┘
```

2. Phase 1 — ETL Pipeline (Power Query)

Step 1.1: Data Ingestion (`SourceData`)

- **Import Source Data**: Navigate to **Home → Get Data → Excel Workbook**. Choose the spreadsheet and select **Transform Data**.
- **Preserve Base Query**: Ensure the raw imported query is named `SourceData`. Keep this untouched as the single source of truth for upstream transformations.

Step 1.2: Sales Query ETL

- **Duplicate Base Query**: Right-click `SourceData` → **Duplicate**. Rename the query to `Sales`.
- **Column Selection**: Select `Salesperson` and columns `January` through `December`. Right-click any selected column header → **Remove Other Columns** *(avoids breakage when raw source schemas shift)*.
- **Unpivot Monthly Attributes**: Select `Salesperson` → **Transform → Unpivot Columns ▼ → Unpivot Other Columns**. Rename `Attribute` → `Month` and `Value` → `Sales`.
- **Construct Composite Key (`MergeKey`)**: Go to **Add Column → Custom Column** named `MergeKey`.

```
Text.Clean(Text.Trim([Salesperson])) & "|" & Text.Clean(Text.Trim([Month]))
```

Step 1.3: Profit Query ETL

- **Duplicate Base Query**: Right-click `SourceData` → **Duplicate**. Rename the query to `Profit`.
- **Column Selection**: Select `Salesperson` and columns `January Profits` through `December Profits`. Right-click → **Remove Other Columns**.
- **Unpivot Monthly Attributes**: Select `Salesperson` → **Transform** → **Unpivot Other Columns**.
- **Text Cleaning**: Select `Attribute` → **Transform** → **Replace Values**. Find ` Profits` *(with leading space)* and Replace With nothing.
- **Rename & Key Generation**: Rename `Attribute` → `Month` and `Value` → `Profit`. Add Custom Column `MergeKey` using the same M formula as above.

Step 1.4: Fact Table Consolidation (`FactSales`)

- **Execute Merge**: Select `Sales` query → **Home** → **Merge Queries** *(Do NOT use Merge as New)*. Select `MergeKey` in `Sales` (Top) and `MergeKey` in `Profit` (Bottom). Set **Join Kind** to `Left Outer`.
- **Data Integrity Audit**: Confirm the status bar reads **96 of 96 rows matched** (or 100% match rate relative to dataset size).
- **Expand Merged Entity**: Click the Expand Icon on the merged `Profit` column. Uncheck `Salesperson`, `Month`, `MergeKey`, and *Use original column name as prefix*. Select **only** **`Profit`**.
- **Rename & Set Data Types**: Rename query to `FactSales` and set explicit data types:

FIELD NAME	TARGET DATA TYPE
`Salesperson`	Text
`Month`	Text
`Sales`	Fixed Decimal Number / Currency
`Profit`	Decimal Number
`MergeKey`	Text

Step 1.5: Temporal Key Engineering

- **Create `Month Number` Column**: Add Custom Column `Month Number`:

```
if [Month] = "January" then 1
else if [Month] = "February" then 2
else if [Month] = "March" then 3
else if [Month] = "April" then 4
else if [Month] = "May" then 5
else if [Month] = "June" then 6
else if [Month] = "July" then 7
else if [Month] = "August" then 8
else if [Month] = "September" then 9
else if [Month] = "October" then 10
else if [Month] = "November" then 11
else if [Month] = "December" then 12
else null
```

- **Create `Date` Column**: Add Custom Column `Date` using `#date(2025, [Month Number], 1)` and set type to **Date**.
- Apply changes: **Home** → **Close & Apply**.

3. Phase 2 — Data Modeling & Date Dimension

Step 2.1: DAX Date Dimension (`DateTable`)

Navigate to **Data View** → **Modeling** → **New Table** and execute:

```
DateTable =
ADDCOLUMNS (
    CALENDAR (
        DATE ( 2025, 1, 1 ),
        DATE ( 2025, 12, 31 )
    ),
    "Year", YEAR ( [Date] ),
    "Month Number", MONTH ( [Date] ),
    "Month", FORMAT ( [Date], "MMMM" ),
    "Year Month", FORMAT ( [Date], "YYYY-MM" )
)
```

Step 2.2: Sort Order & Metadata Configuration

- **Sort Attribute**: Select `DateTable[Month]` → **Column Tools** → **Sort by Column** → **`Month Number`**.
- **Mark as Date Table**: Select `DateTable` → **Table Tools** → **Mark as Date Table** → **Select `Date` column**.

Step 2.3: Relationship Configuration

In **Model View**, establish the primary relationship:

- **From**: `DateTable[Date]` (1) → **To**: `FactSales[Date]` (*)
- **Cardinality**: One to Many (`1:*`) | **Cross Filter Direction**: Single | **Active**: True

> **Modeling Best Practice**
> Disable auto-detected relationships between intermediate queries (`SourceData`, `Profit`) to prevent dynamic filter ambiguity or circular dependency risks.

4. Phase 3 — DAX Measure Engineering

Core Financial KPIs

```
Total Sales = SUM ( FactSales[Sales] )
```

```
Total Profit = SUM ( FactSales[Profit] )
```

```
Profit Margin = DIVIDE ( [Total Profit], [Total Sales], 0 )
```

Time-Scoped Aggregations (Period Windowing)

```
Sales First 3 Months =
CALCULATE (
    [Total Sales],
    REMOVEFILTERS ( DateTable[Month] ),
    DateTable[Month Number] <= 3
)
```

```
Sales Last 3 Months =
CALCULATE (
    [Total Sales],
    REMOVEFILTERS ( DateTable[Month] ),
    DateTable[Month Number] >= 10
)
```

```
Profit First 3 Months =
CALCULATE (
    [Total Profit],
    REMOVEFILTERS ( DateTable[Month] ),
    DateTable[Month Number] <= 3
)
```

```
Profit Last 3 Months =
CALCULATE (
    [Total Profit],
    REMOVEFILTERS ( DateTable[Month] ),
    DateTable[Month Number] >= 10
)
```

Scoped Averages

```
Average Sales First 3 Months =
CALCULATE (
    AVERAGE ( FactSales[Sales] ),
    REMOVEFILTERS ( DateTable[Month] ),
    DateTable[Month Number] <= 3
)
```

```
Average Sales Last 3 Months =
CALCULATE (
    AVERAGE ( FactSales[Sales] ),
    REMOVEFILTERS ( DateTable[Month] ),
    DateTable[Month Number] >= 10
)
```

```
Average Profit First 3 Months =
CALCULATE (
    AVERAGE ( FactSales[Profit] ),
    REMOVEFILTERS ( DateTable[Month] ),
    DateTable[Month Number] <= 3
)
```

```
Average Profit Last 3 Months =
CALCULATE (
    AVERAGE ( FactSales[Profit] ),
    REMOVEFILTERS ( DateTable[Month] ),
    DateTable[Month Number] >= 10
)
```

5. Phase 4 — Visual Layer & UX Configuration

- **Slicers**:
  - **Salesperson Slicer**: Derived from `FactSales[Salesperson]`.
  - **Month Slicer**: Derived from `DateTable[Month]` *(ensures context flows across the model via relationship).*
- **KPI Cards**: Displays for `[Total Sales]`, `[Total Profit]`, and `[Profit Margin]`. Set Display Units to Thousands ($K) with 0 decimal places.
- **Analytical Matrix Visual**:
  - **Rows**: `FactSales[Salesperson]`
  - **Columns**: `DateTable[Month]`
  - **Values**: `[Total Sales]`, `[Total Profit]`

6. Phase 5 — Quality Assurance & Pitfall Matrix

COMMON ERROR / PITFALL	ROOT CAUSE	SOLUTION / PREVENTION
`0 of 96 rows matched` in Merge Step	Mismatched data types or hidden whitespace in key strings.	Apply `Text.Clean(Text.Trim())` on components prior to creating `MergeKey`.
`Null` values in Profit column	`MergeKey` text mismatch or missing records in source query.	Audit individual `MergeKey` values in both queries prior to expanding table.
Total Sales shows unexpected high value (e.g. $6M+)	Aggregating entire dataset without active Slicer context.	Select an individual entity in the `Salesperson` slicer to verify unit filter context.
Months sort alphabetically (April, Aug, Dec...)	Default string sort behavior on text columns.	Set **Sort by Column** on `DateTable[Month]` targeting `DateTable[Month Number]`.
Slicers fail to filter matrix metrics	Using dimension fields directly from fact tables instead of lookup tables.	Use `DateTable[Month]` for slicers, bound through active `1:*` relationship to `FactSales`.

7. Implementation Roadmap Checklist

✓ **Checkpoint 1 (Ingestion)**: Load Excel file and establish `SourceData`.

✓ **Checkpoint 2 (Transformation)**: Generate clean `Sales` and `Profit` queries with standard `MergeKey` fields.

✓ **Checkpoint 3 (Fact Table)**: Merge `Sales` & `Profit` into `FactSales`, expand metrics, verify row count consistency.

- ✓ **Checkpoint 4 (Dates)**: Add `Month Number` and `Date` fields; close Power Query editor.
- ✓ **Checkpoint 5 (Modeling)**: Create DAX `DateTable`, configure sort order, mark as date table, establish `1:*` relationship.
- ✓ **Checkpoint 6 (DAX Engine)**: Build core measures (`Total Sales`, `Total Profit`, `Profit Margin`) and scoped window measures.
- ✓ **Checkpoint 7 (Reporting)**: Assemble interactive Slicers, KPI Cards, and Sales Matrix visual layer.